PROJECT REPORT

Flight Delay Prediction

NYC Flights Dataset (2013)

Binary Classification Using Scheduled Flight Information

Team	T029
Section	E
Prediction cutoff	6:00 PM on the previous day
Dataset	NYC Flights, 2013

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Academic Project Report

Abstract

This project predicts whether a flight will arrive at least 15 minutes late using the NYC flights dataset from 2013. The task is binary classification, with predictions assumed to be made by 6:00 PM on the previous day. To better approximate future deployment, data is split chronologically into training (January–August), validation (September–October), and testing (November–December). Logistic Regression and XGBoost are compared, with a prior-probability Dummy Classifier as a baseline. Exploratory analysis examines class balance and delay-rate variation by scheduled hour, weekday, and carrier. Average Precision is the primary model-selection metric. Logistic Regression achieved validation Average Precision of 0.2348 and ROC-AUC of 0.6446, slightly above XGBoost and above a constant-score baseline. A threshold of 0.2611 was selected on validation predictions to optimize F1-score. On the held-out test period, Logistic Regression achieved Average Precision 0.3359, ROC-AUC 0.6070, precision 0.3415, recall 0.4456, F1-score 0.3867, and accuracy 0.6349. These results indicate moderate predictive signal and must be interpreted in light of temporal variation, feature-availability assumptions, and the lack of weather data.

Keywords: flight delay, binary classification, Logistic Regression, XGBoost, Average Precision, temporal validation, data leakage.

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Chapter 1

Introduction

Flight delays create uncertainty for passengers and operational challenges for airlines and airports. A model that estimates delay risk in advance may help identify flights with elevated delay probability. Its usefulness depends on both predictive performance and whether inputs are genuinely available at prediction time. This project uses NYC flights from 2013 and assumes a prediction cutoff of 6:00 PM on the day before departure.

1.1 Objectives

- Define a binary target for arrival delay.
- Explore delay prevalence and temporal variation.
- Build preprocessing for scheduled-flight features.
- Compare an interpretable linear model with a tree-based model.
- Select the model and threshold on validation data.
- Evaluate on a later held-out test period.

Chapter 2

Problem Formulation and Dataset

2.1 Target Definition

The binary target is

$$y = \begin{cases} 1, & \text{if } \texttt{arr_delay} \geq 15 \text{ minutes}, \\ 0, & \text{if } \texttt{arr_delay} < 15 \text{ minutes}. \end{cases}$$

The raw dataset contains 336,776 rows and 21 columns. Excluding 9,430 rows with missing `arr_delay` leaves 327,346 labelled flights, of which approximately 24.47% are delayed. This setup predicts delay among flights with an observed arrival-delay label; it does not separately predict cancellations or diversions.

2.2 Prediction-Time Assumption

The intended cutoff is 6:00 PM on the previous day. Every feature must be available by this cutoff. A column's presence in the dataset does not prove availability at prediction time; this must be checked feature by feature.

2.3 Features and Leakage Controls

Inputs include categorical `carrier`, `origin`, `dest_grouped`, and `flight`; numeric `distance`, scheduled departure/arrival minute features, and `scheduled_origin_hour_count`; and cyclical hour/calendar encodings where used. The schedule-count feature is not actual congestion, and its availability depends on when the schedule is known. The categorical `flight` feature should be checked for generalization to future or uncommon identifiers.

Prohibited predictor columns are `dep_time`, `dep_delay`, `arr_time`, `air_time`, `arr_delay`, `id`, `name`, `tailnum`, `y`, `date`, and `split`. Actual departure/arrival observations must not be used for a prediction made the previous evening. Imputation, scaling, and categorical encoding must be fitted on training data only and then applied to validation and test

data.

Table 2.1: Feature groups and intended role

Feature group	**Examples**	**Purpose / check**
Categorical schedule fields	Carrier, origin, destination group, flight number	Represent airline, route, and scheduled service differences; verify handling of rare and unseen categories.
Numeric schedule fields	Distance, scheduled departure/arrival minute	Represent route length and scheduled timing.
Cyclical time features	Sine/cosine of hour, weekday, day of year	Represent periodic time patterns without treating the end and start of a cycle as far apart.
Schedule count	Scheduled flights at the same origin-hour	Proxy for scheduled activity, not measured congestion; construction must include only schedules available at the cutoff.

Chapter 3

Exploratory Data Analysis

The raw dataset contains 336,776 flights. After excluding rows without an observed arrival-delay label, 327,346 flights remain. The positive class is defined as an arrival delay of at least 15 minutes; its prevalence is 24.47%. This class distribution motivates the use of Average Precision alongside ROC-AUC rather than relying on accuracy alone.

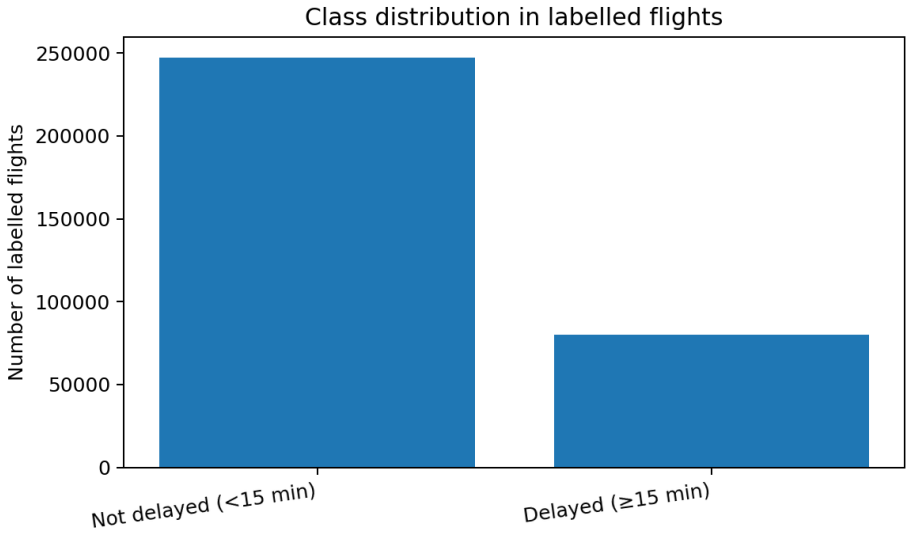


Figure 3.1: Class distribution among flights with an observed arrival-delay label.

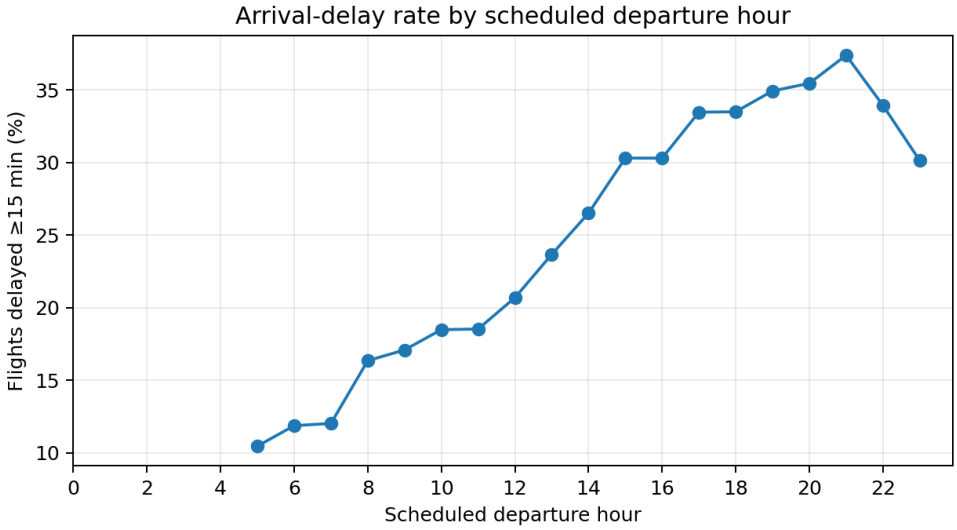


Figure 3.2: Observed arrival-delay rate by scheduled departure hour. This is descriptive analysis, not a causal effect.

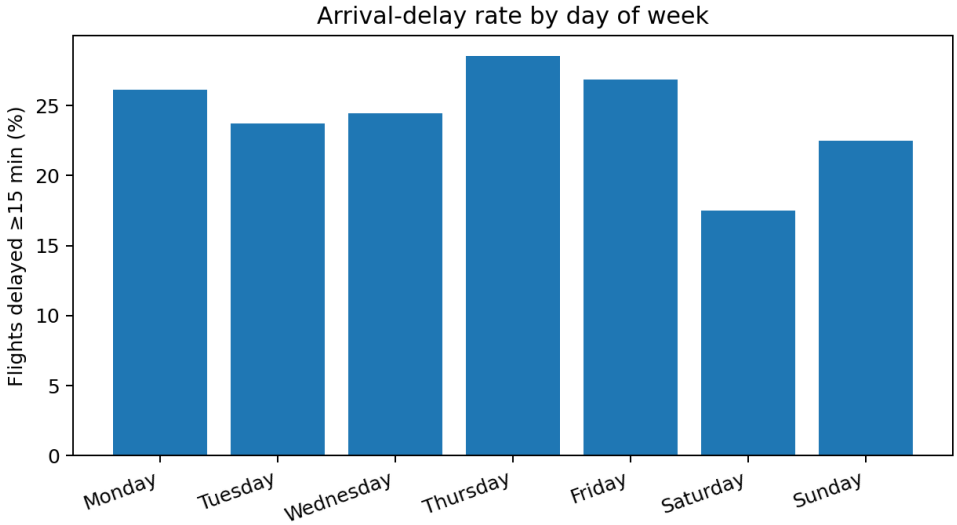


Figure 3.3: Observed arrival-delay rate by day of week. Differences may reflect seasonality, schedule composition, or operational conditions.

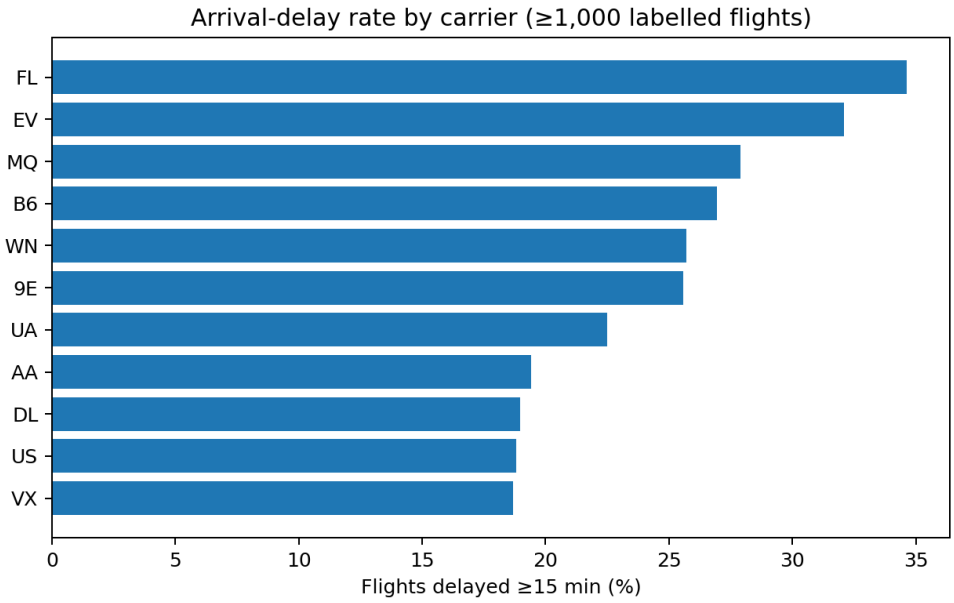


Figure 3.4: Observed arrival-delay rate by carrier, restricted to carriers with at least 1,000 labelled flights to reduce instability from very small groups.

The chronological partitions also have different delay rates:

Table 3.1: Delay prevalence by chronological split

Split	**Period**	**Delay rate**
Training	January–August 2013	26.37%
Validation	September–October 2013	15.70%
Testing	November–December 2013	25.83%

The validation period is substantially calmer than the training and test periods. This is evidence of temporal variation, but the rates alone do not identify its cause. The EDA also shows that delay rates vary by scheduled hour, weekday, and carrier. These are descriptive associations for this dataset and year, not universal rules or causal claims. Weather data is not included, which limits the model's ability to anticipate weather-driven disruption.

Chapter 4

Theoretical Framework and Rigor

4.1 Formal Problem Definition

Let each eligible flight be represented by a feature vector x_i constructed only from information assumed to be available by the prediction cutoff. The binary outcome is

$$y_i = \begin{cases} 1, & \text{if } \texttt{arr_delay}_i \geq 15 \text{ minutes}, \\ 0, & \text{otherwise.} \end{cases}$$

Rows with missing arrival delay are excluded because their binary outcome cannot be determined from this definition. The learning task is to estimate a conditional probability

$$\hat{p}(x_i) \approx P(Y_i = 1 \mid X_i = x_i,\ \mathcal{I}_{t_0}),$$

where $\mathcal{I}_{t_0}$ denotes the information legitimately available at the stated cutoff t_0 (6:00 PM on the previous day). This notation makes the operational constraint explicit: a feature can be statistically predictive and still be invalid if it would not be known at prediction time.

4.2 Research Question and Hypotheses

The central question is whether scheduled-flight attributes and calendar/time features contain useful information for ranking flights by the risk of arriving at least 15 minutes late. The model comparison is guided by the following hypotheses:

- $\mathbf{H}_1$ **(predictive signal):** The selected features provide some ranking information beyond a constant-probability classifier.
- $\mathbf{H}_2$ **(model comparison):** Logistic Regression and XGBoost differ in validation ranking performance, measured primarily by Average Precision (AP).
- $\mathbf{H}_3$ **(temporal generalization):** A model selected using earlier training data and a

later validation period can be evaluated more realistically on a still-later, untouched test period than with a random split.

These are empirical hypotheses, not guaranteed outcomes. In particular, H_1 should only be described as supported after the Dummy Classifier's measured validation AP is compared with the learned models' AP. The available results show that Logistic Regression has slightly higher validation AP than XGBoost; they do not establish that the difference is statistically significant.

4.3 Why the Chronological Split Matters

Flight operations, schedules, seasons, and disruption patterns can change over time. A random split can distribute near-contemporaneous observations across training and evaluation sets and thereby produce an optimistic estimate for a future-prediction task. This project therefore uses January–August for training, September–October for validation, and November–December for testing. The validation set supports model selection and threshold selection; the test set is reserved for the final evaluation.

This design better reflects temporal deployment, but it does not eliminate all uncertainty. It evaluates one calendar year and one particular test window. The validation delay prevalence (15.70%) differs from the training prevalence (26.37%) and test prevalence (25.83%), indicating temporal variation in the outcome distribution. Consequently, validation metrics and an F1-optimized threshold may not transfer unchanged to every future period.

4.4 Model Assumptions and Mathematical Basis

4.4.1 Logistic Regression

For a feature vector x, Logistic Regression models the probability of delay as

$$\hat{p}(x) = \sigma(\beta_0 + \beta^\top x) = \frac{1}{1 + \exp[-(\beta_0 + \beta^\top x)]}.$$

Equivalently, it assumes a linear relationship between the predictors and the log-odds:

$$\log \frac{P(Y = 1 \mid X = x)}{1 - P(Y = 1 \mid X = x)} = \beta_0 + \beta^\top x.$$

This model is a useful baseline because it is relatively simple and its coefficients can be inspected after appropriate preprocessing. Its main limitation is that the log-odds are linear in the encoded features unless nonlinear transformations or interactions are added. Regularization, if enabled in the implementation, can reduce coefficient magnitude and

help control overfitting; the exact penalty and hyperparameters should be stated from the notebook rather than assumed.

4.4.2 XGBoost

XGBoost builds an additive ensemble of decision trees. In simplified form, its prediction score is

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \qquad f_k \in \mathcal{F},$$

where each f_k is a tree from the permitted tree-function space $\mathcal{F}$. Training optimizes a regularized objective,

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k),$$

where ℓ measures prediction error and Ω penalizes model complexity. Trees can represent nonlinear effects and feature interactions without requiring them to be specified manually, but greater flexibility can also increase overfitting risk and sensitivity to hyperparameters. The validation comparison is therefore necessary; theoretical flexibility alone is not evidence of better performance.

4.5 Class Imbalance and Metric Choice

The positive class is a flight arriving at least 15 minutes late. Its observed prevalence in the labelled dataset is approximately 24.47%, so accuracy alone can obscure performance on delayed flights. Average Precision summarizes precision over recall levels and is used as the primary ranking metric. ROC-AUC measures the ability to rank a randomly selected positive example above a randomly selected negative example across thresholds. Both are threshold-independent ranking metrics, but neither directly specifies an operational decision threshold.

At a chosen threshold, precision, recall, and F1-score describe a particular trade-off:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN},$$

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

The threshold of 0.2611 was selected using validation predictions to optimize F1 and then applied unchanged to the test set. This avoids choosing the threshold on test outcomes. However, maximizing F1 is a project decision, not a universal optimum: an airline or passenger-facing system may assign different costs to missed delays and false alarms.

4.6 Leakage Control and Reproducibility

A valid predictive evaluation requires that information derived from the target or observed after the prediction cutoff not enter the feature set. Accordingly, realized operational variables such as departure delay, arrival time, air time, and arrival delay are excluded from predictors. The target, row identifier, and split label are also excluded. Preprocessing parameters—including imputers, scalers, and category encoders—must be fitted on training data only and then applied to validation and test data.

The cutoff-time claim requires particular care. A scheduled-flight count feature is not automatically a measure of actual congestion, and its availability by 6:00 PM on the previous day must be verified from its construction. Likewise, a flight identifier encoded as a categorical predictor may be sparse or unseen in later periods and may not generalize reliably. These are data-provenance and deployment questions, not matters that can be resolved by model scores alone.

4.7 Interpretation of the Evidence

The validation results are Logistic Regression AP = 0.2348 and ROC-AUC = 0.6446, compared with XGBoost AP = 0.2301 and ROC-AUC = 0.6368. The differences favor Logistic Regression but are small; without repeated temporal evaluations or uncertainty estimates, they should not be interpreted as proof of statistically meaningful superiority. On the held-out test period, Logistic Regression achieves AP = 0.3359 and ROC-AUC = 0.6070. The fact that test AP is higher while test ROC-AUC is lower is not inherently contradictory: the metrics summarize different ranking properties and are also affected by the test-period class distribution and score ordering.

The results support a cautious conclusion: the features provide a potentially useful but limited signal, and performance varies across time. A stronger claim about improvement over a naive baseline requires reporting the actual Dummy Classifier validation AP. Stronger claims about stability would require additional years or rolling-origin evaluation, and calibrated probability claims would require calibration analysis.

Chapter 5

Methodology

5.1 Chronological Split

Training uses January–August 2013, validation uses September–October 2013, and testing uses November–December 2013. This better reflects prediction on future flights than a random split. The test set must remain separate from feature/model selection and threshold tuning.

5.2 Preprocessing

Numeric variables may require imputation and scaling; categorical variables require encoding. All learned preprocessing operations should be fitted only on training data. The same fitted transformations are then applied consistently to validation and test partitions.

5.3 Models

Dummy Classifier: The `prior` strategy uses the training class distribution rather than learning meaningful relationships. Its actual validation AP must be checked before claiming improvement over the baseline.

Logistic Regression: An interpretable linear classifier that estimates delay probability and provides a useful baseline.

XGBoost: A gradient-boosted tree model capable of representing nonlinear relationships and feature interactions.

5.4 Validation Model Selection

Average Precision (AP) is the primary selection metric, with ROC-AUC as a complementary ranking metric.

Table 5.1: Validation performance

Model	**Average Precision**	**ROC-AUC**	**AP lift**
Dummy Classifier (constant score)	0.1570	0.5000	1.00
Logistic Regression	0.2348	0.6446	1.50
XGBoost	0.2301	0.6368	1.47

The constant-score Dummy Classifier has AP equal to the validation positive-class prevalence (0.1570) and ROC-AUC of 0.5000. Logistic Regression therefore improves validation AP by about 0.0778 absolute points, or approximately 49.6% relative to this baseline. Logistic Regression also narrowly exceeds XGBoost on the reported validation AP and ROC-AUC. The difference between the two learned models is small and does not establish statistically significant or universal superiority. These figures must be checked against the final executed notebook before submission.

Chapter 6

Evaluation Metrics

6.1 Average Precision

Average Precision summarizes precision across recall levels and is the primary metric because delayed flights are the minority class. Higher AP indicates better ranking of positive examples relative to negative examples. Lift relative to positive-class prevalence is

$$\text{AP Lift} = \frac{\text{Average Precision}}{\text{Positive-Class Prevalence}}.$$

Both quantities must be computed on the same evaluation split.

6.2 ROC-AUC

ROC-AUC measures ranking discrimination across thresholds. A value of 0.5 corresponds to random ranking and 1.0 to perfect ranking. It complements AP but can appear relatively optimistic with imbalanced classes.

6.3 Threshold-Dependent Metrics

For true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN):

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad \text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Precision measures the share of predicted delays that are actually delayed; recall measures the share of actual delays detected. F1 is their harmonic mean. Accuracy is supplementary because class imbalance can make it misleading. A confusion matrix separates false alarms from missed delays.

Chapter 7

Threshold Selection and Test Results

7.1 Threshold Selection

The selected probability threshold is **0.2611**, chosen on validation predictions to optimize F1-score and then applied unchanged to test predictions. A flight is classified as delayed when its predicted probability is at least this threshold. The threshold is a decision rule, not a quality metric; a lower threshold generally flags more flights and may raise recall while lowering precision. Validation-based selection avoids tuning on the test set.

7.2 Held-Out Test Performance

Table 7.1: Test metrics at threshold 0.2611

Metric	**Test result**
Average Precision	0.3359
AP lift over test prevalence	1.30
ROC-AUC	0.6070
Precision	0.3415
Recall	0.4456
F1-score	0.3867
Accuracy	0.6349

The model identifies approximately 44.56% of actual delayed flights. Of flights flagged as delayed, 34.15% are actually delayed according to the target definition. This indicates some predictive signal, but performance remains moderate: more than half of actual delays are missed at this threshold, and many flagged flights are not delayed.

Test AP is higher than validation AP, while test ROC-AUC is lower. This does not by itself prove leakage or better generalization. Different delay prevalence and period composition should be considered.

Chapter 8

Discussion

Logistic Regression slightly outperformed XGBoost on validation metrics. The available results do not establish why; possibilities include relatively simple relationships in the features, limited signal for a more complex model, or temporal effects. Chronological evaluation is appropriate for forecasting, but one year and one held-out period cannot establish performance across other years. The validation delay rate (15.70%) differs notably from training (26.37%) and testing (25.83%), so validation results may not represent all future periods. Finally, absence of obvious target leakage does not guarantee a feature was available at the cutoff. Feature provenance and timing need independent verification.

Chapter 9

Limitations

- Delay prevalence changes substantially across time partitions.
- Weather and other real-time operational signals are absent.
- Availability of every feature at the prediction cutoff requires verification.
- Logistic Regression may miss nonlinear patterns and interactions.
- An F1-optimized threshold may not suit every operational objective.
- Flights without observed arrival-delay labels are excluded; cancellations and diversions are not separately modelled.
- The baseline validation AP must be checked before claiming a quantified improvement over it.

Chapter 10

Conclusion

This project formulates arrival-delay prediction as binary classification with a 15-minute threshold and a prediction cutoff of 6:00 PM on the previous day. Chronological splits provide a more realistic evaluation than random splitting. Logistic Regression was selected because its validation AP (0.2348) and ROC-AUC (0.6446) were slightly higher than XGBoost's. A threshold of 0.2611 selected on validation data was applied to the test set. Test AP 0.3359, ROC-AUC 0.6070, precision 0.3415, recall 0.4456, F1 0.3867, and accuracy 0.6349 suggest moderate predictive capability, not reliable detection of every delay. Interpretation should account for temporal variation, class imbalance, feature availability, and the limited feature set.

Appendix A

Reproducibility Checklist

1. Verify the raw CSV has 336,776 rows and 21 columns.
2. Verify 9,430 missing arrival-delay labels and 327,346 labelled rows.
3. Exclude all prohibited columns from predictors.
4. Fit preprocessing on training data only.
5. Verify every feature is available by the previous-day cutoff.
6. Record Dummy Classifier validation AP from the actual run.
7. Select model and threshold on validation data only.
8. Apply threshold 0.2611 unchanged to test predictions.
9. Tie every reported metric to its correct split.

Implementation and Reproducibility Notes

The submitted notebook should be treated as the executable record of the experiment. Before final submission, confirm that the values in this report match one single final run. Record the Python, scikit-learn, and XGBoost versions; the random seed where applicable; the exact preprocessing pipeline; and the model hyperparameters. Save the final validation comparison, selected threshold, test predictions, and confusion matrix as generated outputs. Do not combine metrics from different notebook versions or runs.

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